

## **High-level proposal: Tendency-Driven AI Agents (Group 5)**

### **Concept overview**

We propose an AI agent framework where each agent's behavior emerges from a stable but adaptable set of base tendencies (similar to human-like drives). Instead of hardcoding behaviors as scripts, agents evaluate possible actions by predicting the effects of those actions and combining those effects with their internal tendencies. Actions are chosen based on the highest overall "fit" between what the agent expects will happen and what the agent is biased to care about.

This approach is intentionally designed so that behavior is not a single reward maximization problem. Instead, agents act as a messy combination of competing tendencies (e.g., curiosity, affiliation, fear/avoidance, task-focus). Over time, tendencies can shift based on experience, producing emergent patterns such as avoidance/anxiety-like behavior (high affiliation desire paired with high fear sensitivity) or exploration-driven personalities.

### **Goals**

Produce believable agent variety. Different agents should behave differently in the same environment because their internal tendency weights differ.

Support emergent behavior. Complex social and behavioral outcomes should arise from repeated interactions over time, rather than from hand-authored scripts.

Keep implementation feasible. The initial system should work using simple numeric scoring and heuristic effect prediction, without requiring heavy machine learning. More advanced learning or evolutionary approaches may be added later if feasible.

Provide a clean interface for integration. Agents should follow a consistent evaluation pipeline that can operate independently of the specific world module used.

### **World context**

The project may take place in a simple, pre-industrial world that may range from primitive to loosely medieval in structure. The environment may include small settlements, work areas, social spaces, shared resources, and basic infrastructure. The exact level of development will depend on feasibility and the direction chosen by the overall project.

Rather than focusing on historical accuracy, the world is designed to support interaction, cooperation, construction, and adaptation. Agents should be able to act meaningfully whether the world is minimally developed or somewhat structured.

This flexibility allows the same agent model to support evolutionary behavior and long-term adaptation, while also allowing a simpler, more constrained world if needed to ensure reliable integration and timely completion.

The world provides a small, interpretable set of actions such as building, helping, trading, avoiding, exploring, and socializing. These actions map naturally to internal tendencies and allow emergent social dynamics to appear within short simulation runs.

Alternatively, we could change the model to interact with a player's posts in a world that uses social media more. The ai would be able to look at tags from the post (examples being location in the game world, time, people) and based on that react to the post using emojis. As the player progress through the world, the agents will be able to react to the player's post dynamically which could be used to influence the main goal of the game (such as gaining followers or reaching like goals)

## **Core model**

The agent model is built around the idea that all agents share a common, human-like internal structure rather than fundamentally different personalities. Just as humans broadly share similar emotional and motivational tendencies, all agents in the simulation operate using the same core set of tendencies.

Behavioral diversity does not come from agents having different kinds of drives, but from how these shared tendencies interact with randomness, context, and accumulated experience over time.

### **A) Tendencies (shared emotional sensitivities)**

Each agent maintains a fixed-length vector of tendency weights. These tendencies represent sensitivities to certain classes of outcomes, not explicit emotions or commands. All agents use the same tendency dimensions, with the same semantic meaning.

Examples of core tendencies:

affiliation\_weight

Sensitivity to social bonding, cooperation, and proximity to others.

curiosity\_weight

Sensitivity to novelty, exploration, and new experiences.

fear\_weight

Sensitivity to risk, threat, uncertainty, or potential harm.

task\_weight

Sensitivity to productive or goal-oriented outcomes such as building, crafting, or maintenance.

comfort\_weight

Sensitivity to effort, stability, familiarity, and avoidance of unnecessary cost.

These tendencies are present in all agents. Initial weights may vary slightly, but large personality differences are not assumed at the start.

Differences between agents emerge because:

- small initial differences are amplified over time
- agents encounter different sequences of events
- randomness affects action selection and outcomes
- experiences update sensitivities in different ways

This mirrors how humans share emotional architecture but diverge behaviorally.

## **B) Action effect features (interpreting the world)**

When an agent considers a possible action, it estimates a vector of expected effects. This vector represents the agent's prediction of what the action will cause in the world and social environment.

Action effect features are neutral descriptions of consequences, not emotions.

Examples of effect dimensions:

bonding\_gain

Expected improvement in social relationships or trust.

novelty\_gain

Expected exposure to new places, agents, or information.

risk\_exposure

Expected likelihood of harm, loss, or conflict.

energy\_cost

Expected effort, time, or resource expenditure.

progress\_gain

Expected contribution toward construction, crafting, or task completion.

conflict\_generated

Expected social tension or harm caused.

stability\_gain

Expected increase in predictability or maintenance of familiar structures.

Effect values are estimated using simple heuristics based on distance, tags, memory, and basic world rules. Precision is not required; consistency is more important than accuracy.

## **C) Decision rule (tendencies shaping behavior)**

For each decision step, the agent follows the same evaluation process:

1. Generate a small set of candidate actions based on the current world state.

2. For each action, construct an effect feature vector describing predicted outcomes.
3. Combine the effect features with the agent's tendency weights using a weighted sum or dot product.
4. Select the action with the highest score, optionally with small randomness to allow exploration.

Because all agents share the same tendency structure, different behavior arises from:

- different weighting strengths
- different past experiences influencing perception
- stochastic variation in selection
- The same action may be chosen by different agents for different underlying reasons.

#### **D) Tendencies as emotions (emergent interpretation)**

Tendencies are best understood as emotional sensitivities, not explicit emotional states.

For example:

- Fear is not a binary state, but a sensitivity that amplifies negative outcomes.
- Curiosity is not an instruction to explore, but a bias toward novelty.
- Affiliation is not a rule to socialize, but a weighting that increases the value of social effects

Emotional-like states emerge from how these sensitivities interact over time with predicted and actual outcomes.

For instance, anxiety-like behavior can emerge when:

- affiliation sensitivity remains high
- fear sensitivity increases due to repeated negative social outcomes
- novelty and exploration are increasingly penalized
- No explicit “anxiety” variable is required. The behavior emerges from conflicting sensitivities.

The model supports gradual adaptation of tendencies based on experience.

After an action is taken:

- the agent compares predicted effects to actual outcomes
- discrepancies influence sensitivity updates

General patterns:

- repeated negative outcomes increase fear sensitivity and avoidance

- repeated positive outcomes increase trust and affiliation sensitivity
- surprising outcomes may increase generalization or caution
- These updates are small and gradual, preserving stability while allowing long-term behavioral change.

If feasible, evolutionary tuning may also be applied

- multiple agents are evaluated across episodes
- fitness is measured using balanced objectives (social bonds, productivity, survival, harm avoidance)
- mutation or crossover adjusts initial tendency weights or learning rates

## **Integration with Group 5 classes**

FeatureVector

Used to represent both tendency vectors and action effect vectors with a fixed numeric layout.

AnnotationSet

Stores symbolic memory about entities, locations, and experiences (for example: trusted, dangerous, helpful, harmful).

TagManager

Provides fast retrieval of relevant entities or objects using tags and tag combinations.

FunctionSet

Organizes modular evaluation steps such as candidate generation, effect estimation, scoring, and post-action updates.

RobinHoodMap

Used internally for high-performance storage of memory, tag indices, and agent state.

## **Example agent loop**

Each simulation tick:

1. The agent observes the local world state.
2. Relevant entities and objects are identified using TagManager.
3. A small set of candidate actions is generated.
4. An effect feature vector is estimated for each action.
5. Actions are scored using the agent's shared tendency weights.
6. The selected action is executed.
7. Memory is updated, and tendencies may be adjusted based on outcomes.

Over time, the agent's personality and behavioral style emerge from this repeated evaluation process. The kinds of actions an agent tends to take, the situations it avoids or seeks out, and the social roles it naturally falls into are not explicitly defined ahead of time. Instead, they arise from how the agent's shared tendencies interact with the world, randomness, and accumulated experience.

Depending on feasibility and project direction, we may also opt for a simpler implementation. In this version, tendencies may be hardcoded per agent, or randomly initialized and kept fixed over time, without learning or adaptation. This still allows us to observe how different initial tendency configurations lead to different agent behaviors in the same environment, even if tendencies themselves do not evolve.

This flexibility allows the project to demonstrate meaningful behavioral variation at multiple levels of complexity, while ensuring reliable integration with the rest of the system.

### **Scope and implementation flexibility**

This proposal describes a high-level direction and conceptual framework for the AI Agents module rather than a fixed or guaranteed implementation. The final structure and complexity of the agent system may evolve as the project progresses.

The degree to which tendency-driven behavior, learning, or evolutionary mechanisms are implemented will depend on:

- overall project scope and time constraints,
- how the World modules are ultimately designed,
- integration requirements with other groups,
- and what proves most reliable and maintainable in practice.

While the full model emphasizes shared tendencies, emergent behavior, and experience-driven adaptation, we may choose to implement a simpler subset of this framework if it better supports integration and stability. For example, tendencies may remain fixed, learning may be limited or omitted, or the action evaluation process may be simplified.

Because agent behavior depends heavily on how the world exposes actions, states, and interactions, close collaboration with the World teams is essential. The agent design is intended to adapt to the world rather than impose strict requirements on it.

The priority is to deliver a robust, clearly integrated system that demonstrates meaningful agent behavior within the final simulation. Additional complexity will only be added if it aligns with the overall project direction and does not compromise integration, testing, or reliability.

